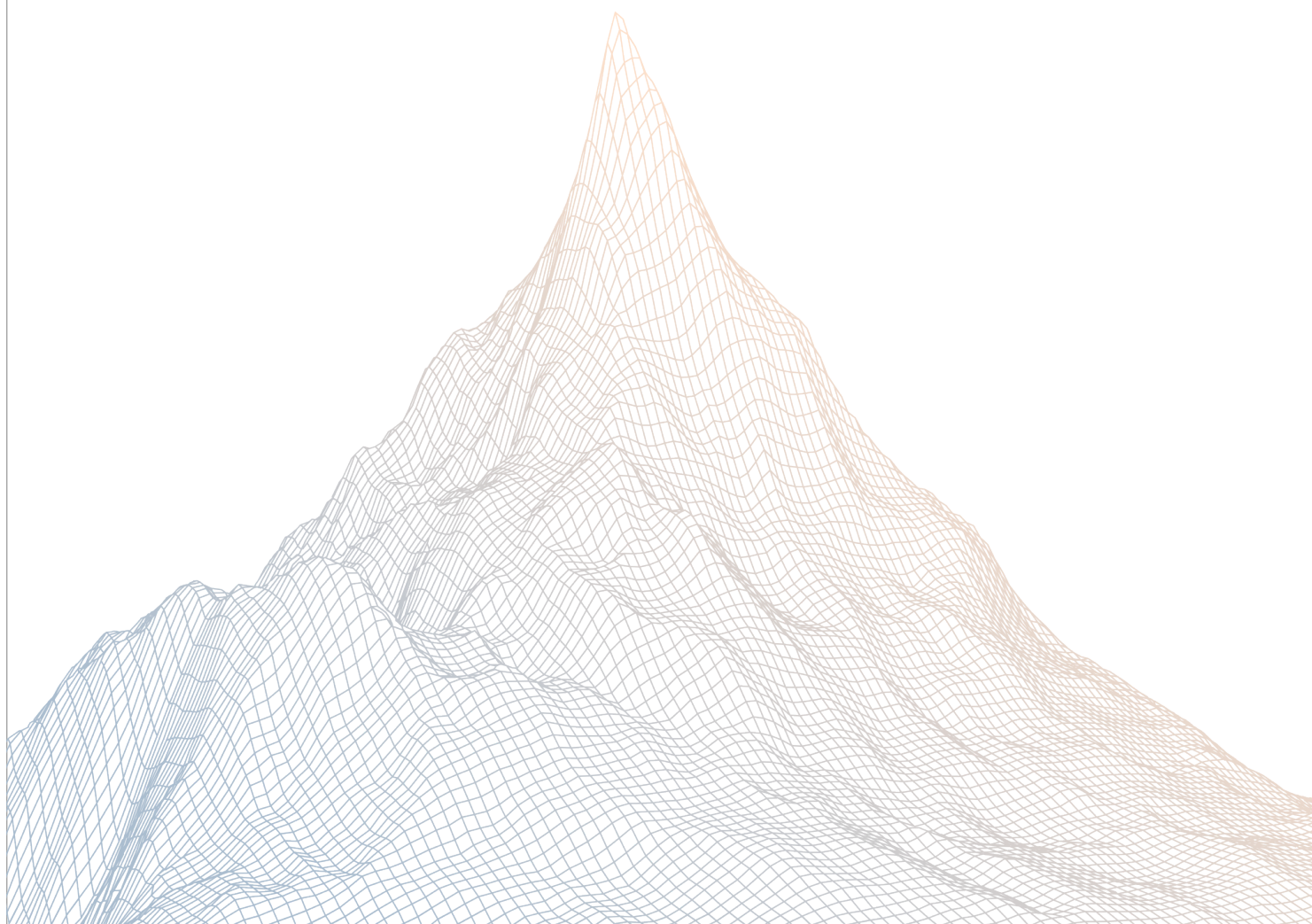


Berachain

Smart Contract Security Assessment

VERSION 1.1



AUDIT DATES:

May 4th to May 6th, 2026

AUDITED BY:

Matte
ether_sky

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Introduction

1.1 About Zenith

Zenith assembles auditors with proven track records: finding critical vulnerabilities in public audit competitions.

Our audits are carried out by a curated team of the industry's top-performing security researchers, selected for your specific codebase, security needs, and budget.

Learn more about us at <https://zenith.security>.

1.2 Disclaimer

This report reflects an analysis conducted within a defined scope and time frame, based on provided materials and documentation. It does not encompass all possible vulnerabilities and should not be considered exhaustive.

The review and accompanying report are presented on an "as-is" and "as-available" basis, without any express or implied warranties.

Furthermore, this report neither endorses any specific project or team nor assures the complete security of the project.

1.3 Risk Classification

SEVERITY LEVEL	IMPACT: HIGH	IMPACT: MEDIUM	IMPACT: LOW
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

2

Executive Summary

2.1 About Berachain

Berachain is a high-performance EVM-Identical Layer 1 blockchain utilizing Proof-of-Liquidity (PoL) and built on top of the modular EVM-focused consensus client framework BeaconKit.

2.2 Scope

The engagement involved a review of the following targets:

Target	contracts-internal
Repository	https://github.com/berachain/contracts-internal/pull/107
Commit Hash	06ef847f679d56b825d7ae351ac515fb18bd3a9a
Files	Diff from 340ff9cdf99386acf44754d43f99afd2a3dd2474 src/base/**/* src/pol/**/*

2.3 Audit Timeline

May 4, 2026	Audit start
May 6, 2026	Audit end
May 8, 2026	Report published

2.4 Issues Found

SEVERITY	COUNT
Critical Risk	0
High Risk	0
Medium Risk	0
Low Risk	1
Informational	0
Total Issues	1

3

Findings Summary

ID	Description	Status
L-1	WBERA can still be sent to the distributor even while the emission token remains BGT	Acknowledged

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Findings

4.1 Low Risk

A total of 1 low risk findings were identified.

[L-1] WBERA can still be sent to the distributor even while the emission token remains BGT

SEVERITY: Low

IMPACT: High

STATUS: Acknowledged

LIKELIHOOD: Low

Target

- [BlockRewardController.sol](#)

Description:

In the distributor, the current emission token is BGT, and the default admin can update it to WBERA.

```
function setEmissionToken() external onlyRole(DEFAULT_ADMIN_ROLE) {  
    emissionToken = WBERA_ADDRESS;  
    emit EmissionTokenSet(emissionToken);  
}
```

However, in the BlockRewardController, WBERA is sent to the distributor regardless of whether the emission token has been updated to WBERA or not.

```
function _handleMinting(address receiver, uint256 amount) internal {  
    if (address(this).balance < amount) {  
        bgt.mint(address(this), amount);  
        bgt.redeem(address(this), amount);  
    }  
  
    wbera.deposit{ value: amount }();  
    wbera.transfer(receiver, amount);  
}
```

Recommendations:

Ensure the `emission` token is updated immediately after the contract upgrade, or have the `BlockRewardController` transfer the correct `emission` token to the distributor.

Berachain: Acknowledged. The contract upgrade will occur as an atomic transaction with a `multisig`. Specifically, we will use the `upgradeToAndCall()` function to upgrade the contract and call the method for the `Distributor`.